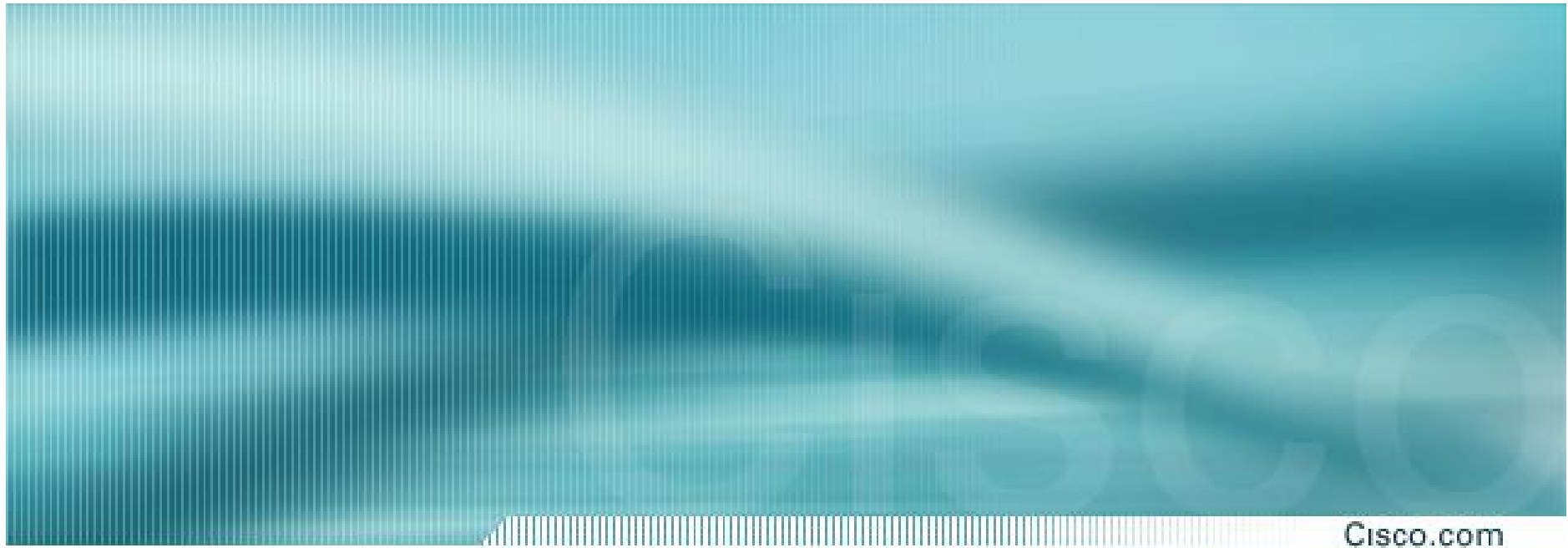




IP Addressing and Subnetting

Objectives

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Upon completion, you will be able to:

- Discuss the **Types** of Network Addressing
- Explain the **Form** of an IP **Address**
 - **Network ID**
 - **Host ID**
- Discuss the **Classes** of IP Addresses
- Understand the Function of the **Mask**

Let's Talk About Addressing!

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- **Types of Addressing:**
 - **Layer 2 – MAC Addresses (Media Access Control)**
0134.2345.12AB A **MAC** Address
0134.23 Vendor Code
45.12AB Serial Number
 - **Layer 3 – Logical Addresses (IPv4 or IPX)**
- **Assignment of IP Addresses:**
 - **Static Addresses – assigned by an Administrator**
 - **Dynamic Addresses – DHCP**
 - **“Hierarchical” vs. “Flat” Addressing Schemes**

Can You Count in Binary?

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**We are Very Familiar with our
Decimal System...**

0 1 2 3 4 5 6 7 8 9...10 11 12 13...

But,

**We Need to Become Familiar with the
Binary System...only 0's and 1's**

0000 1 10 11 100 101 110 111 1000 1001...

0 1 2 3 4 5 6 7 8 9

Basics of An IPv4 Address

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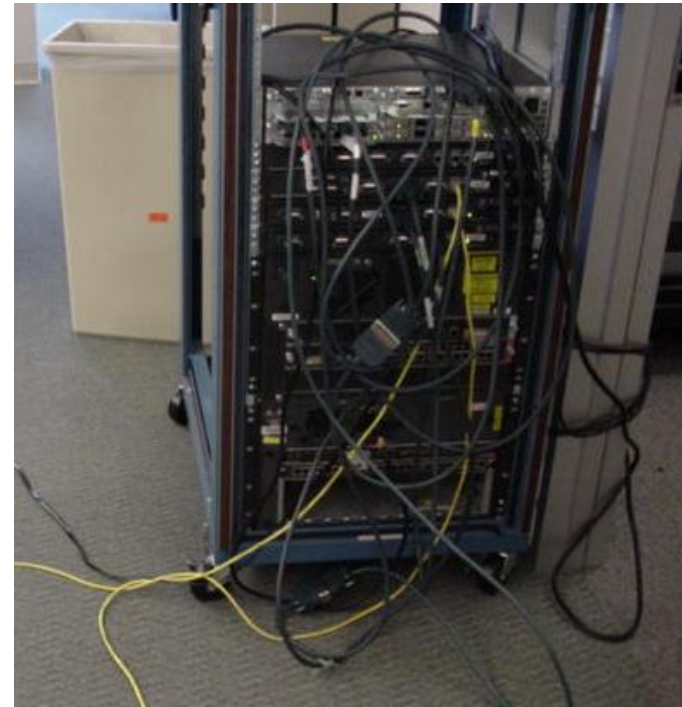
- Layer 3 (L3) Logical IP Addresses are comprised of 4 Octets, separated by a .
- The **Decimal** form looks like this:

176.223.14.127

- The **Binary** form looks like:

128 64 32 16 8 4 2 1

10110000.11011111.00001110.01111111



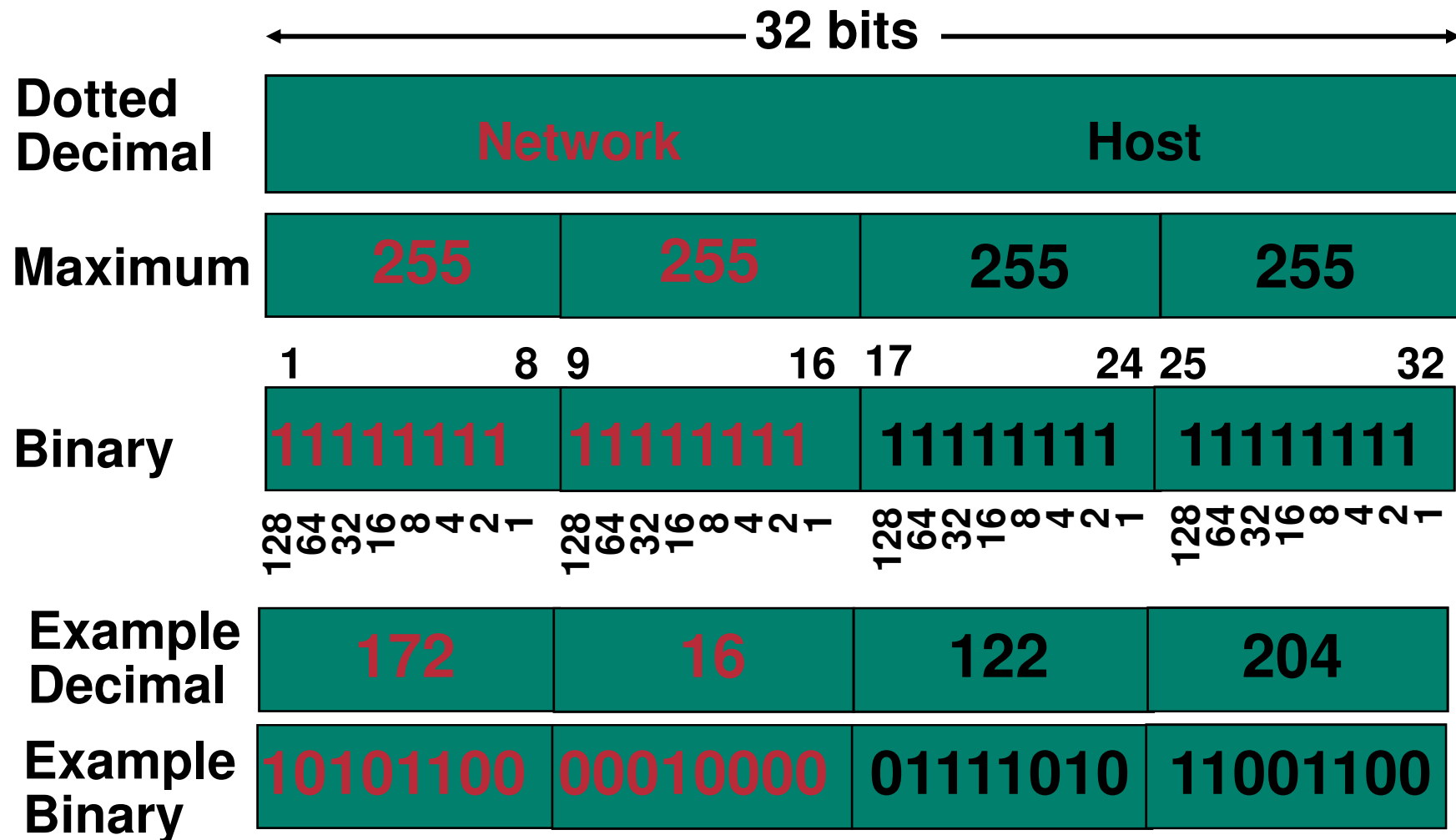
Basics of An IPv4 Address

- Each of the 4 Octets has **8 Bits**
- Each of these Bits has a “Binary Value”
- Each Bit can only be a **One** or a **Zero**
- Let’ s Look at One of the Octets – **8 Bits**



Each of these 8 bits has a distinct value, that starts at “1” from the right side and moving to the left, doubles each time to 2, 4, 8, 16, 32, 64, and finally 128, as shown above.

IPv4 Addressing



IPv4 Address Classes

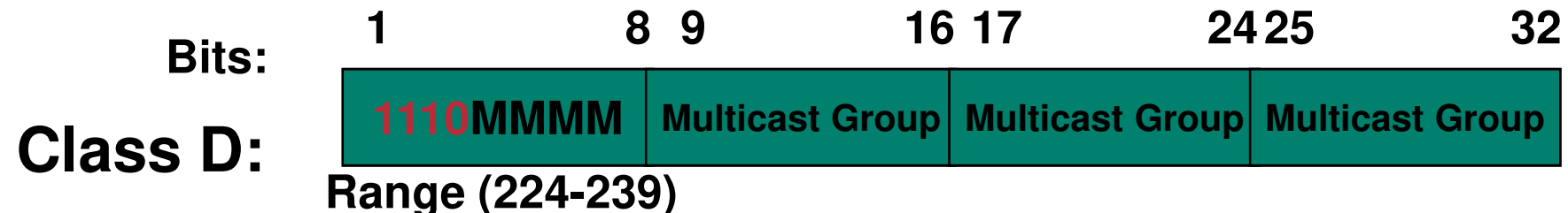
- **Class A:**

8 bits	8 bits	8 bits	8 bits
Network	Host	Host	Host
- **Class B:**

Network	Network	Host	Host
---------	---------	------	------
- **Class C:**

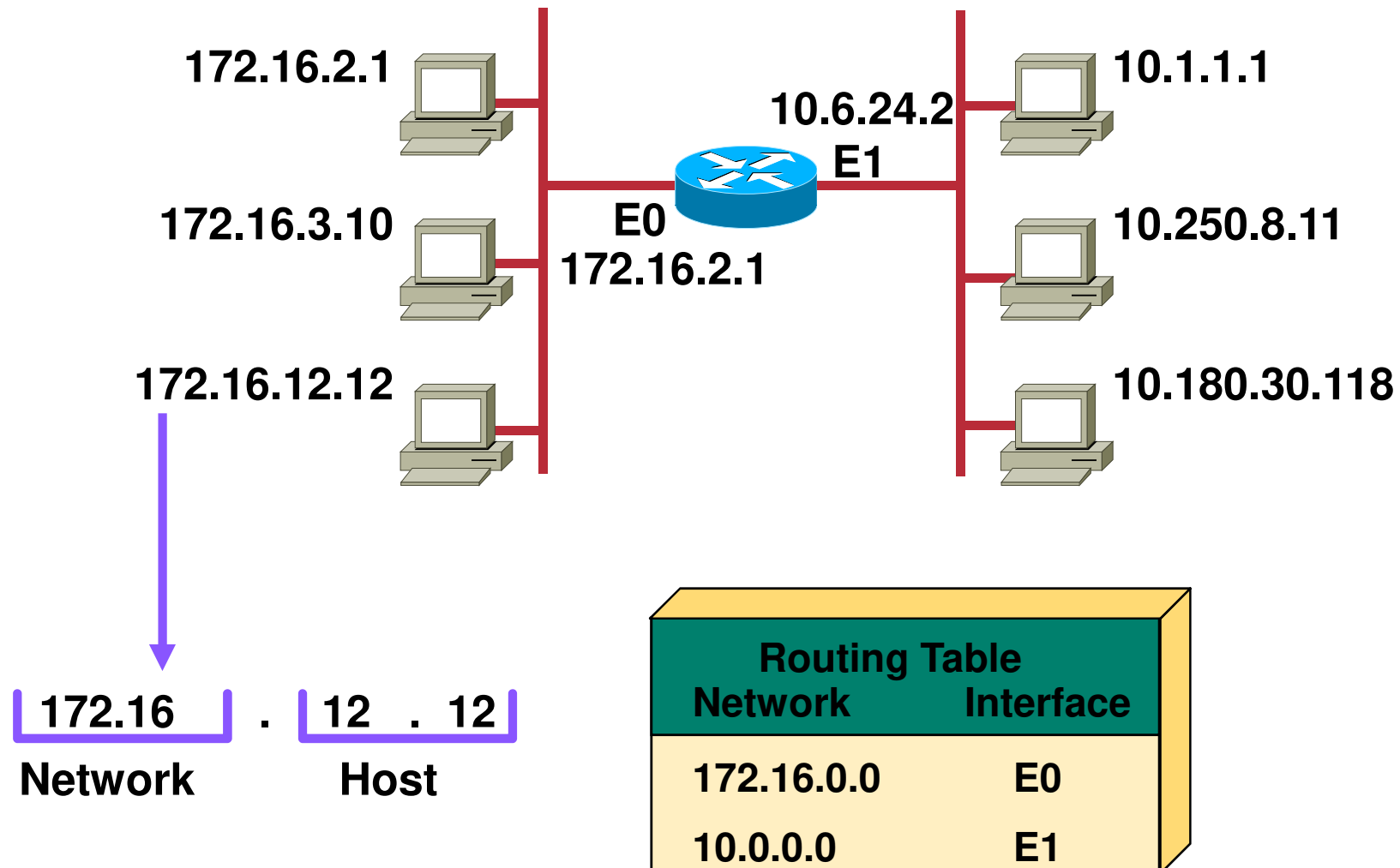
Network	Network	Network	Host
---------	---------	---------	------
- **Class D:** Multicast
- **Class E:** Research

IPv4 Address Classes



Host Addresses

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Determining Available Host Addresses

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Network		Host															
172	16	0 0															

10101100 00010000

16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 N

00000000	00000000	1
00000000	00000001	2
00000000	00000011	3
⋮	⋮	⋮
11111111	11111101	65534
11111111	11111110	65535
11111111	11111111	65536

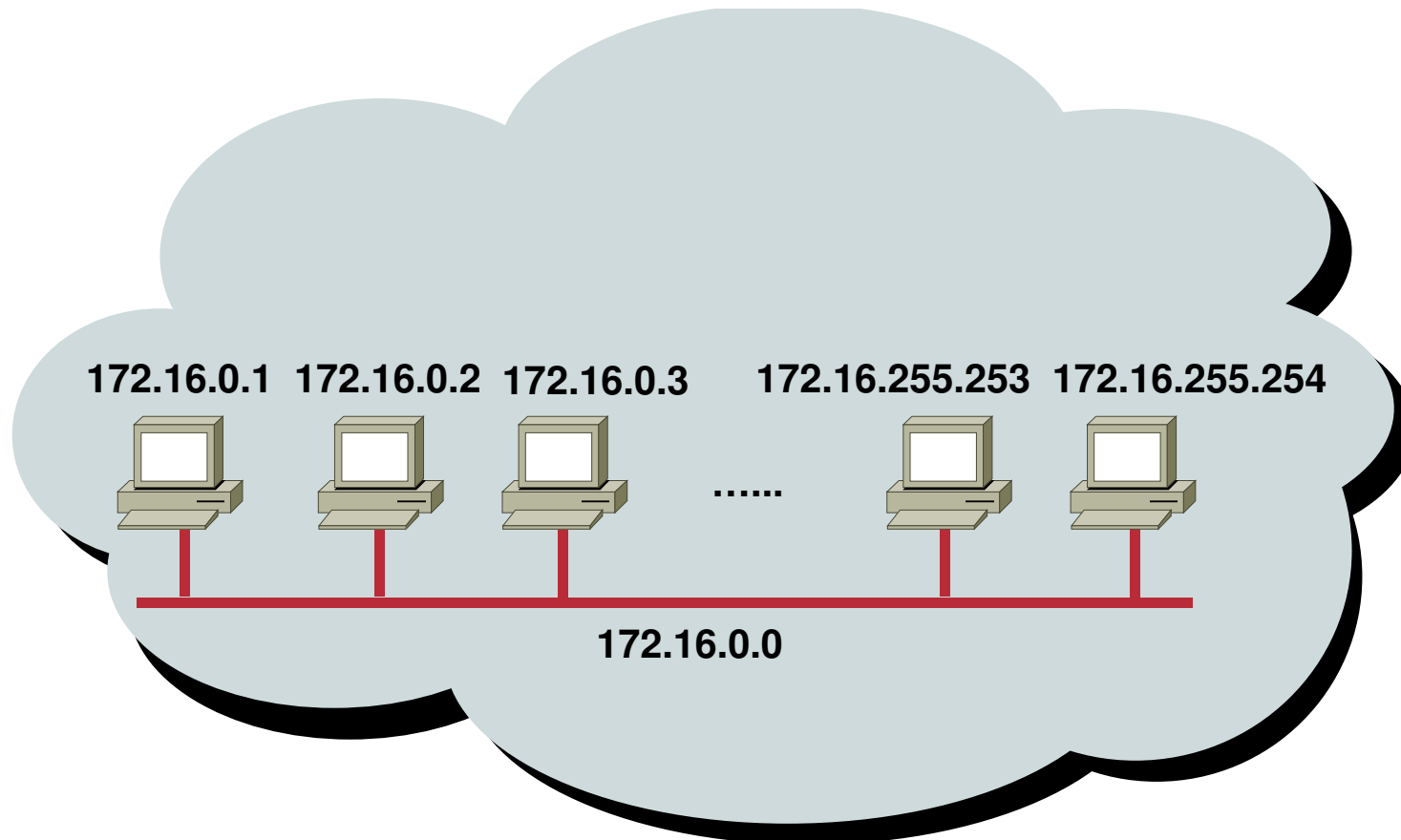
Remember $2^N - 2$
(where N is the
number
of host bits)

$2^N - 2 = 2^{16} - 2 = 65534$

65536	
- 2	
65534	

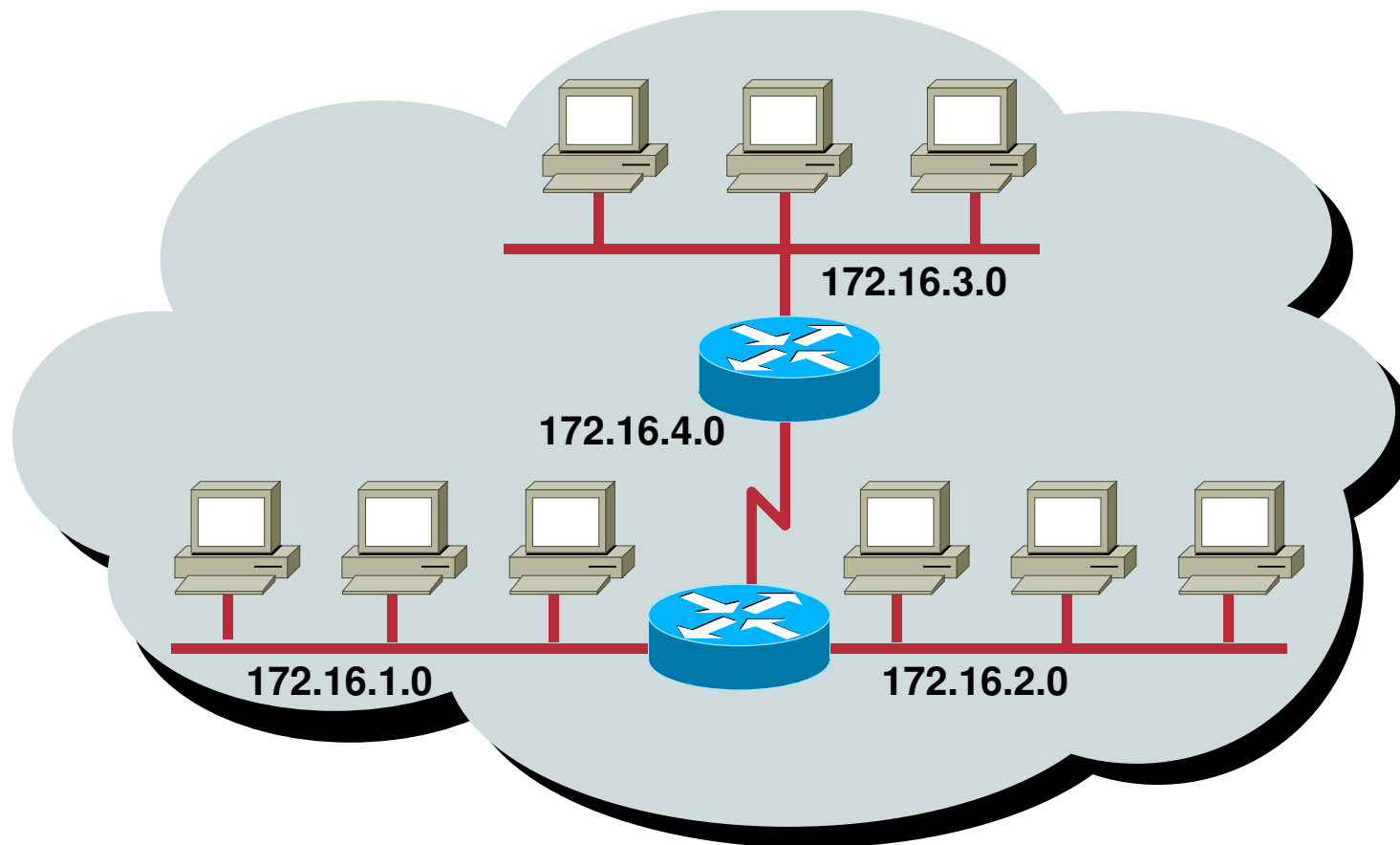
Addressing without Subnets

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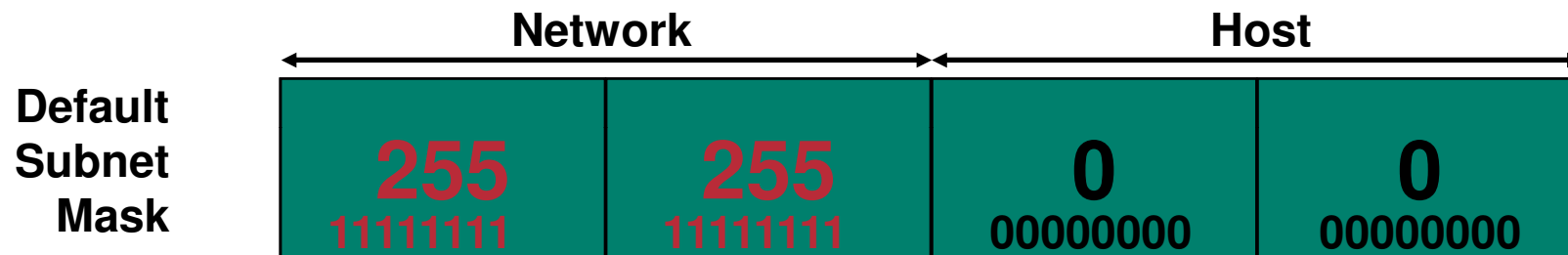
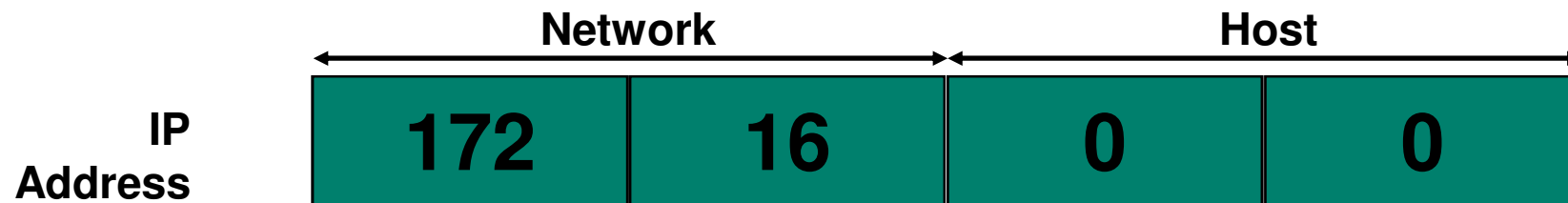
- **Network 172.16.0.0**

Addressing with Subnets

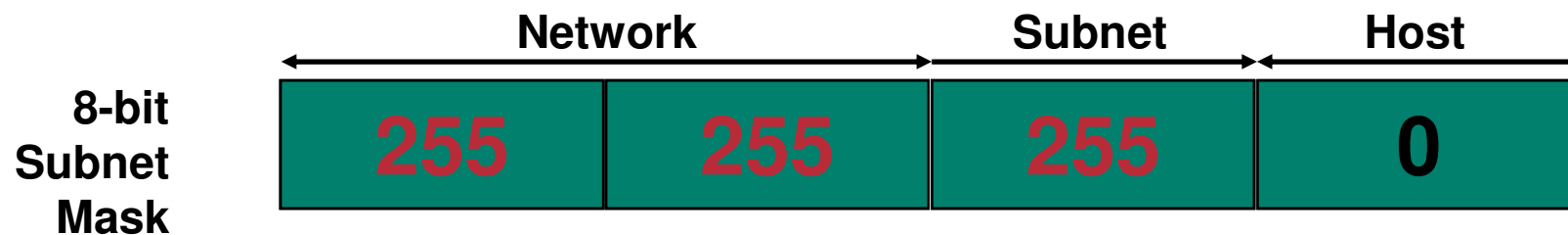


- **Network 172.16.0.0**

Subnet Mask



Also written as “/16” where 16 represents the number of 1s in the mask.



Also written as “/24” where 24 represents the number of 1s in the mask.

Decimal Equivalents of Bit Patterns

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128	64	32	16	8	4	2	1	
↓	↓	↓	↓	↓	↓	↓	↓	
1	0	0	0	0	0	0	0	= 128
1	1	0	0	0	0	0	0	= 192
1	1	1	0	0	0	0	0	= 224
1	1	1	1	0	0	0	0	= 240
1	1	1	1	1	0	0	0	= 248
1	1	1	1	1	1	0	0	= 252
1	1	1	1	1	1	1	0	= 254
1	1	1	1	1	1	1	1	= 255

Know your two's

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- $2^1 = 2$
- $2^2 = 4$
- $2^3 = 8$
- $2^4 = 16$
- $2^5 = 32$
- $2^6 = 64$
- $2^7 = 128$
- $2^8 = 256$
- $2^9 = 512$
- $2^{10} = 1024$
- $2^{11} = 2048$
- $2^{12} = 4096$
- $2^{13} = 8192$
- $2^{14} = 16384$
- $2^{15} = 32768$
- $2^{16} = 65536$

Know Your CIDR Values

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- | | | | |
|-----------------|-----|-------------------|-----|
| • 255.0.0.0 | /8 | • 255.255.224.0 | /19 |
| • 255.128.0.0 | /9 | • 255.255.240.0 | /20 |
| • 255.192.0.0 | /10 | • 255.255.248.0 | /21 |
| • 255.224.0.0 | /11 | • 255.255.252.0 | /22 |
| • 255.240.0.0 | /12 | • 255.255.254.0 | /23 |
| • 255.248.0.0 | /13 | • 255.255.255.0 | /24 |
| • 255.252.0.0 | /14 | • 255.255.255.128 | /25 |
| • 255.254.0.0 | /15 | • 255.255.255.192 | /26 |
| • 255.255.0.0 | /16 | • 255.255.255.224 | /27 |
| • 255.255.128.0 | /17 | • 255.255.255.240 | /28 |
| • 255.255.192.0 | /18 | • 255.255.255.248 | /29 |
| | | • 255.255.255.252 | /30 |

Subnet Mask with Subnets

	Network		Subnet	Host
172.16.2.160	10101100	00010000	00000010	10100000
255.255.255.0	11111111	11111111	11111111	00000000
	10101100	00010000	00000010	00000000

128 192 224 240 248 252 254 255

Network
Number

172	16	2	0
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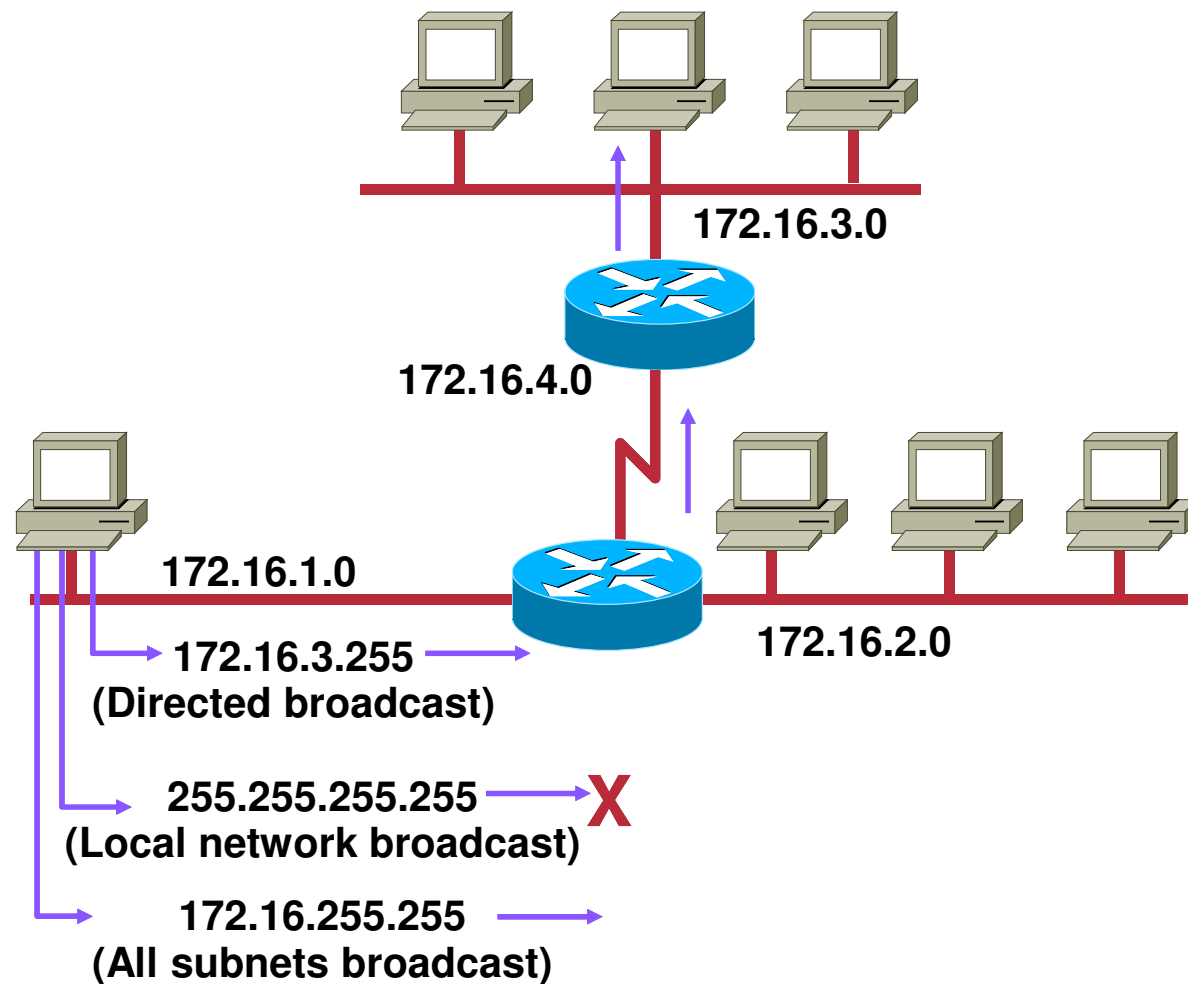
- Network number extended by eight bits
- Without a subnet mask you cannot tell the host address nor the network it resides on!

Subnet Mask with Subnets (cont.)

	Network		Subnet	Host
172.16.2.160	10101100	00010000	00000010	10100000
255.255.255.192	11111111	11111111	11111111	11000000
	10101100	00010000	00000010	10000000
			128 192 224 240 248 252 254 255	128 192 224 240 248 252 254 255
Network Number	172	16	2	128

- Network number extended by ten bits

Broadcast Addresses



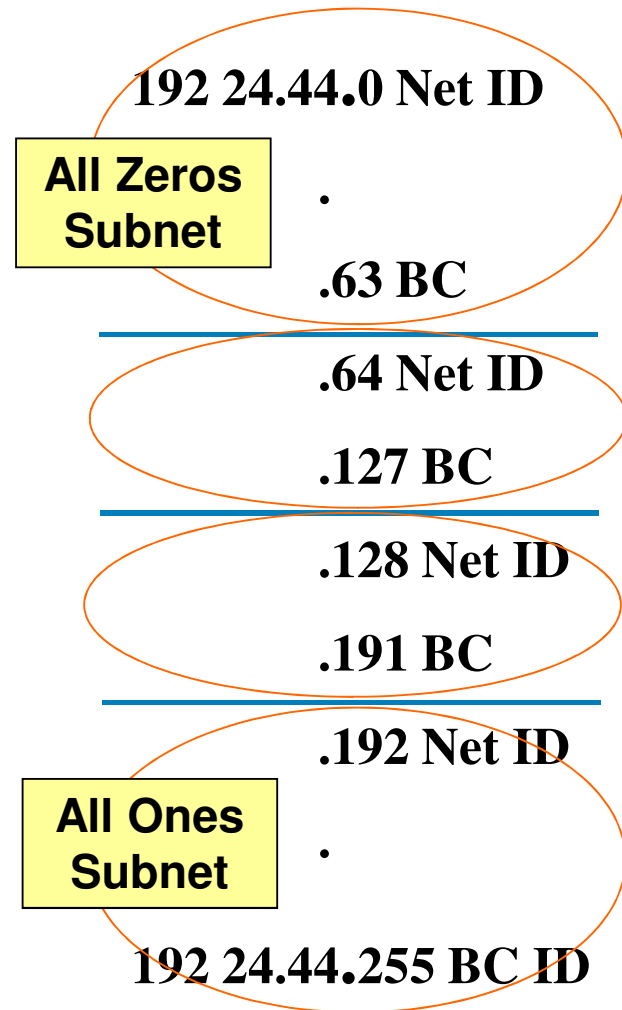
All Zeros and All Ones Subnets

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RFC 1878 states:

"This practice of excluding the "all-zeros subnet" and the "all-ones subnet" is obsolete! Modern software will be able to utilize all definable sub-networks."

Today, the use of subnet zero and the all-ones subnet is generally accepted and most vendors support their use, though, on certain networks (and the **CCNA Exam**), particularly the ones using legacy software, the use of subnet zero and the all-ones subnet can lead to problems.



VLSM - Variable Length Subnet Mask

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